Experiment 5

Report Generation using Relational and Multidimensional Data on the Star Schema in Power BI

Business Intelligence Lab · Unit 2, Session 5 · Source file: exp_5.pbix (Global Superstore 2018 data)

$12.64M	$1.47M	11.6%	25,728	51,279
Total Sales	Total Profit	Profit margin	Orders	Fact rows

What this report covers

This document summarises the Power BI file saved for Experiment 5: its pages and visuals, the star-schema data model behind them, and the figures each report type produces. Every number below was recomputed from the data stored inside the .pbix, using the same fields and measures the visuals use. Where the saved file differs from what the Experiment 5 brief asks for, that is stated plainly in the Findings section so it can be fixed before submission.

Headline results

- **Sales grew every year**, from $2,259,154 in 2014 to $4,299,763 in 2017 (+90%).
- **Technology** is the largest category by sales ($4,744,501) and by profit; **Furniture** has the lowest margin (6.9%).
- **Western Europe** is the most profitable region ($218,423); **Western Asia** and **Western Africa** lose money.
- **38,043** of 51,248 order lines (74%) are profitable; the largest is Tamara Chand's Canon Imageclass 2200 copier at $17,499.95.

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1. Report structure

The file contains four report pages. Names are as saved in Power BI.

Page	Visuals	What it shows
Page 1	4 KPI cards (Total Sales, Total Profit, Orders Sales, Orders Profit), combo chart	Headline KPIs and a combo chart of sales, profit, quantity, shipping cost and margin. Several of these use the raw *Orders* table rather than the star schema.
Page 2	Clustered column, pie, donut, hierarchy slicer	Profit, sales, margin by Country; pie by Customer ID; donut of sales/profit/quantity; Country → City → State slicer on *Orders*. Exploratory from Experiment 4.
Relational Report	Table, Matrix, small Category table	Transaction table (Order ID, Customer, Product, Region, Sales, Profit); matrix on Region → State → City with Total Sales, Total Profit, Quantity; Category × Sales table.
Comparative Analysis	Column, bar, line, 100% stacked bar	Sales by Category; Profit by Region (Top 10 filter); yearly line of Total Sales, Profit and Quantity; 100% stacked bar of Profit Margin/Profit by Sales.

2. Data model

A star schema with one fact table and four dimensions, plus the raw source tables loaded from *Global Superstore 2018.xlsx*.

Table	Role	Rows	Key columns
Fact_sales	Fact	51,279	Order ID, Order Date, Customer ID, Region, Product ID, Sales, Quantity, Discount, Profit
Dim_Customer	Dimension	17,415	Customer ID, Customer Name, Segment
Dim_Product	Dimension	3,788	Product ID, Product Name, Sub-Category, Category
Dim_Region	Dimension	3,828	Region, Country, State, City (hierarchy “Region Hierarchy”)
Dim_Date	Dimension	51,290	Order Date, Year, Quarter, Month Name
Orders / People / Returns	Source	51,290 / – / –	Raw Superstore sheets (24 columns in Orders)

Relationships

From (Fact_sales)	To	Cardinality	Filter direction	Matches brief?
Customer ID	Dim_Customer[Customer ID]	Many to one	Single	Yes
Product ID	Dim_Product[Product ID]	Many to one	Single	Yes
Order Date	Dim_Date[Order Date]	**Many to many**	Both	No – should be many to one
Region	Dim_Region[Region]	**Many to many**	Both	No – should be many to one

Measures

Measure	DAX
Total Sales	SUM(Fact_sales[Sales])
Total Profit	SUM(Fact_sales[Profit])
Profit Margin	DIVIDE(SUM(Fact_sales[Profit]), SUM(Fact_sales[Sales]))
Average Order Value	DIVIDE([Total Sales], DISTINCTCOUNT(Fact_sales[Order ID]))

3. Relational report (transaction table)

A plain table of individual orders with **Order ID, Customer Name, Product Name, Region, Sales, Profit**, built on Fact_sales joined to three dimensions. This is the level of detail an operations team uses to check a specific order.

Applying the brief's steps (sort by Sales high to low, filter Profit > 0) to this table gives **38,043** rows out of 51,248 order lines. The ten largest are shown below.

Order ID	Customer	Product	Region	Sales	Profit
CA-2016-TC20980140-42646	Tamara Chand	Canon Imageclass 2200 Advanced Copier	Central Us	17,499.95	8,399.98
CA-2017-RB19360140-42818	Raymond Buch	Canon Imageclass 2200 Advanced Copier	Western Us	13,999.96	6,719.98
CA-2017-TA21385140-43031	Tom Ashbrook	Canon Imageclass 2200 Advanced Copier	Eastern Us	11,199.97	3,919.99
CA-2017-HL15040140-43057	Hunter Lopez	Canon Imageclass 2200 Advanced Copier	Eastern Us	10,499.97	5,039.99
CA-2016-AB10105140-42722	Adrian Barton	Gbc Ibimaster 500 Manual Proclick Binding System	Central Us	9,892.74	4,946.37
CA-2014-SC20095140-41904	Sanjit Chand	Ibico Epk-21 Electric Binding System	Central Us	9,449.95	4,630.48
US-2016-BS11365140-42477	Bill Shonely	3D Systems Cube Printer, 2Nd Generation, Magenta	Eastern Us	9,099.93	2,365.98
CA-2016-SE20110140-42402	Sanjit Engle	Hp Designjet T520 Inkjet Large Format Printer - 24" Color	Southern Us	8,749.95	2,799.98
CA-2016-CC12370140-42514	Christopher Conant	Canon Imageclass 2200 Advanced Copier	Eastern Us	8,399.98	1,120.00
CA-2014-KL16645140-41844	Ken Lonsdale	High Speed Automatic Electric Letter Opener	Western Us	8,187.65	327.51

Observation. 38,043 rows remain after the Profit > 0 filter (74% of lines); 13,205 loss-making or break-even lines are removed. At the top of the sorted table is **Tamara Chand** buying a **Canon Imageclass 2200 Advanced Copier** (Central US, Sales 17,499.95, Profit 8,399.98). Copiers dominate the top of the list, so a few high-value technology orders account for the biggest single transactions.

Note on the saved file: the table is present with all six fields, but no sort order and no Profit > 0 condition are stored on it, so it currently lists every line. Re-apply both in Power BI before exporting to Excel.

4. Multidimensional report (matrix)

The matrix puts the Region → State → City hierarchy in Rows and Total Sales, Total Profit and Quantity in Values, so one visual summarises the same rows at three levels of detail. Region-level totals (the top level of the matrix) are below.

Region	Total Sales	Total Profit	Quantity
Western Europe	1,731,893	218,423	22,259
Central America	1,223,101	158,982	20,882
Oceania	1,100,146	120,077	12,834
Southeastern Asia	884,079	17,763	11,817
Southern Asia	866,573	159,336	9,109
Eastern Asia	855,059	167,102	8,922
Western Us	725,458	108,418	12,266
Eastern Us	678,500	91,535	10,616
Northern Europe	636,779	83,924	8,177

Region	Total Sales	Total Profit	Quantity
South America	616,562	28,142	11,334
Southern Europe	608,510	70,082	7,868
Central Us	501,240	39,706	8,780
Southern Us	391,722	46,749	6,209
Caribbean	324,281	34,571	6,299
Western Asia	317,107	-53,922	5,441
Eastern Europe	310,033	77,085	3,609
North Africa	233,160	57,826	2,920
Western Africa	173,879	-50,408	3,305
Central Africa	143,630	35,384	1,553
Eastern Africa	127,836	21,893	1,687
Southern Africa	105,192	24,159	1,096
Canada	66,928	17,817	833
Central Asia	19,311	-7,282	465
Total	**12,640,978**	**1,467,362**	**178,281**

Drill-down check (Part 3, step 19)

The brief asks the City-level matrix numbers to match the Experiment 4 Drill page for the same state. Taking **California** (Western US) as the example, the true city-level sales from the source data are small — 81 cities, e.g. Los Angeles 175,851 and San Francisco 112,669. In this model, however, the matrix will show **725,458** (the whole Western US region) against *every* city and state in that region.

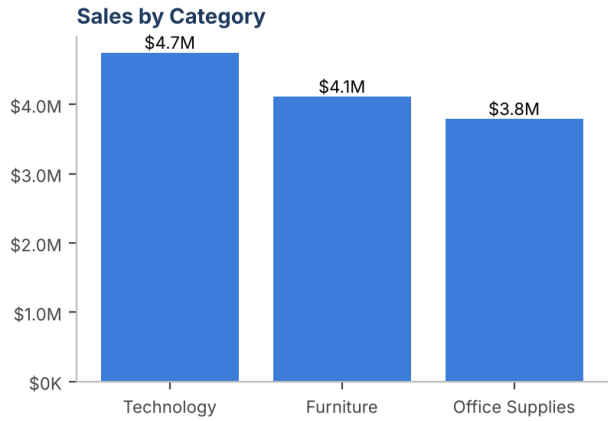
City (California)	True sales (source data)	Matrix shows (model as built)
Los Angeles	175,851	725,458
San Francisco	112,669	725,458
San Diego	47,521	725,458
Anaheim	7,987	725,458
Fresno	7,889	725,458
Sacramento	7,311	725,458

Observation. The matrix and the city-level drill chart will **not** agree. The cause is the relationship between Fact_sales and Dim_Region: it joins on the Region column only, and Region repeats across thousands of State/City rows (many to many). Filtering by a city therefore filters the fact table to its whole region. Region-level numbers are correct; State and City numbers repeat the region total. The model is reliable down to Region, but it cannot yet be trusted for drill-down below it.

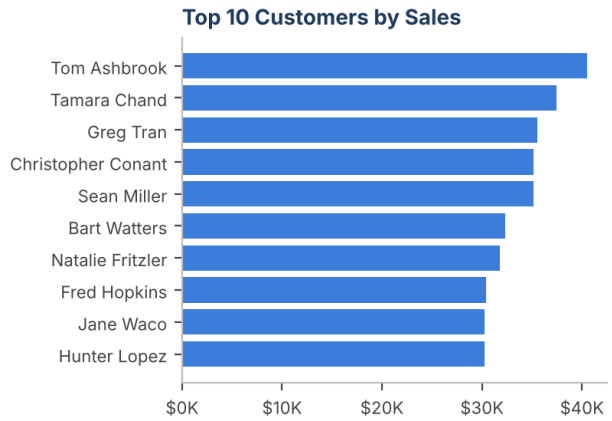
The fix is to give Dim_Region a unique key per City (for example Country-State-City, or a Location ID) and store that key in Fact_sales, so the relationship becomes many to one.

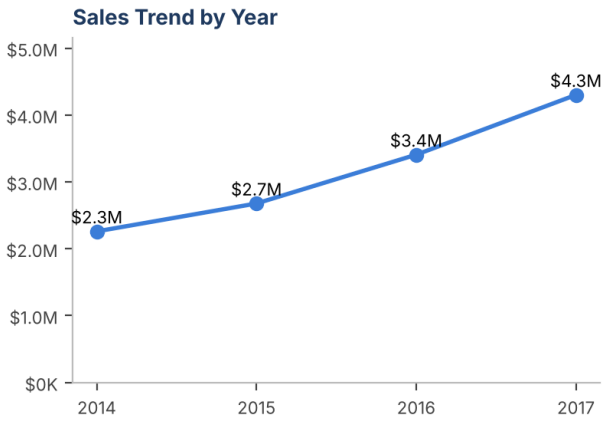
5. Comparative analysis

Four views of the same Fact_sales data. The first, second and fourth are in the saved file; the Top 10 Customers chart is not built yet and is computed here from the model so the page can be completed.









Charts: Sales by Category · Profit by Region (Top 10 by profit, as filtered in the file) · Top 10 Customers by Sales (computed, not yet in file) · Sales Trend by Year.

Observation. The charts show things the transaction list cannot. Technology leads on sales (4.74M) and profit (0.66M), while Furniture sells nearly as much as Office Supplies but earns far less profit (7% margin vs 14%). Western Europe earns the most profit by a wide margin, yet two regions (Western Asia, Western Africa) lose money overall. Sales rose every year, +90% from 2014 to 2017, and the top customers each spend around $30–40K. The Part 2 table, by contrast, only tells us about individual orders; it cannot show category mix, regional ranking, customer concentration or a trend over time.

6. Summary table and interpretation

Report type	Fields / measures used	Key observation	Business use
Relational report	Order ID, Customer Name, Product Name, Region, Sales, Profit	38,043 profitable order lines; largest is Tamara Chand's copier order (17,499.95).	Checking, auditing and resolving individual orders.
Multidimensional report	Region Hierarchy (Region, State, City), Total Sales, Total Profit, Quantity	Region totals are correct (Western Europe highest sales, 1.73M); State/City values repeat the region total because of the many-to-many Region link.	Summarising performance by geography and drilling from region to city.
Comparative analysis	Category, Region, Customer Name, Year, Total Sales	Technology leads; Furniture has thin margin; sales up 90% 2014 to 2017; two loss-making regions.	Spotting trends, rankings and where to focus.

Report type	Fields / measures used	Key observation	Business use
Dashboard	Total Sales, Total Profit, Total Orders, Region, Category, Year	Not in this file (Experiment 6). Page 1 holds the KPI cards it would build on.	Interactive, single-screen monitoring with slicers.

Regional manager, quarterly review (step 25). A regional manager would open the dashboard, set the Region and Year slicers to their territory and quarter, and read totals, category mix and trend at a glance, to see whether the region is on target and which category or customer is pulling it up or down. The relational table is used afterwards, only to look up the specific orders behind a number that looks wrong. The dashboard answers "how are we doing?"; the table answers "which orders caused it?".

7. Findings against the brief

Checked against the Experiment 5 submission checklist. "Done" is in the file, "Partly" is started, "No" still needs doing.

Checklist item	Status	Detail
Four relationships, Many to One	No	Fact_sales→Dim_Date and →Dim_Region are many-to-many, bi-directional. Customer and Product are correct.
Geo_Hierarchy intact	Partly	Hierarchy exists on Dim_Region as "Region Hierarchy" (Region → State → City), not "Geo_Hierarchy".
Relational Report page: table, sorted, filtered	Partly	Table with correct six fields exists; no sort and no Profit > 0 filter saved. Export to Excel cannot be confirmed from the file.
Matrix Report page	Partly	Matrix built with the hierarchy and three values, but it sits on the Relational Report page, not its own "Matrix Report" page.
Comparative Analysis, four titled charts	Partly	Sales by Category, Profit by Region and a yearly line exist. No Top 10 Customers chart. None of the visuals has a title. The line chart mixes Quantity and Profit with Total Sales.
Dashboard (Experiment 6)	No	No dashboard page, cards with slicers or cross-filtering page in the file.
Summary table and interpretation	Done	Completed in section 6 of this report.

Other data-quality points

• **Discount** is stored as a whole number in Fact_sales and Orders, so fractional discounts (e.g. 0.2) have been rounded to 0 or 1. Reload the column as a decimal number.
• Fact_sales has 51,279 rows against 51,290 in Orders, so 11 source rows were dropped when the fact table was built. Total Sales differs by about 1,524.
• Page 1 and Page 2 mix the raw *Orders* table with the star schema, so cards on the same page can show slightly different totals. Use the Fact_sales measures consistently.
• Page names "Page 1", "Page 2" and "Comparative Analysis." (trailing full stop) should be renamed.